

PREDICTING FLIGHT DELAYS

A Machine Learning Approach to Operational Delay Forecasting

Tunisair Operational Data

Luca Tomarelli • Qutibah Hussein

Findings, Recommendations & Future Work

Flight Delays Are a Costly, Cascading Problem

1

Reduced Efficiency

Delays ripple through the day's schedule, disrupting downstream operations.

2

Capital & Repositioning Costs

Aircraft end up out of position for their next scheduled rotation.

3

Crew Reallocation Expense

Duty-time limits force costly crew swaps when delays run long.

4

Eroded Passenger Trust

Severe, memorable delays are what actually reduce future demand.

Our Goal: Predict the delay duration (in minutes) for each flight — enabling proactive contingency planning instead of reactive firefighting.

THE DATA

107,833 Flights, Most On Time — A Few Wildly Not

107,833

Flight Records

64.6%

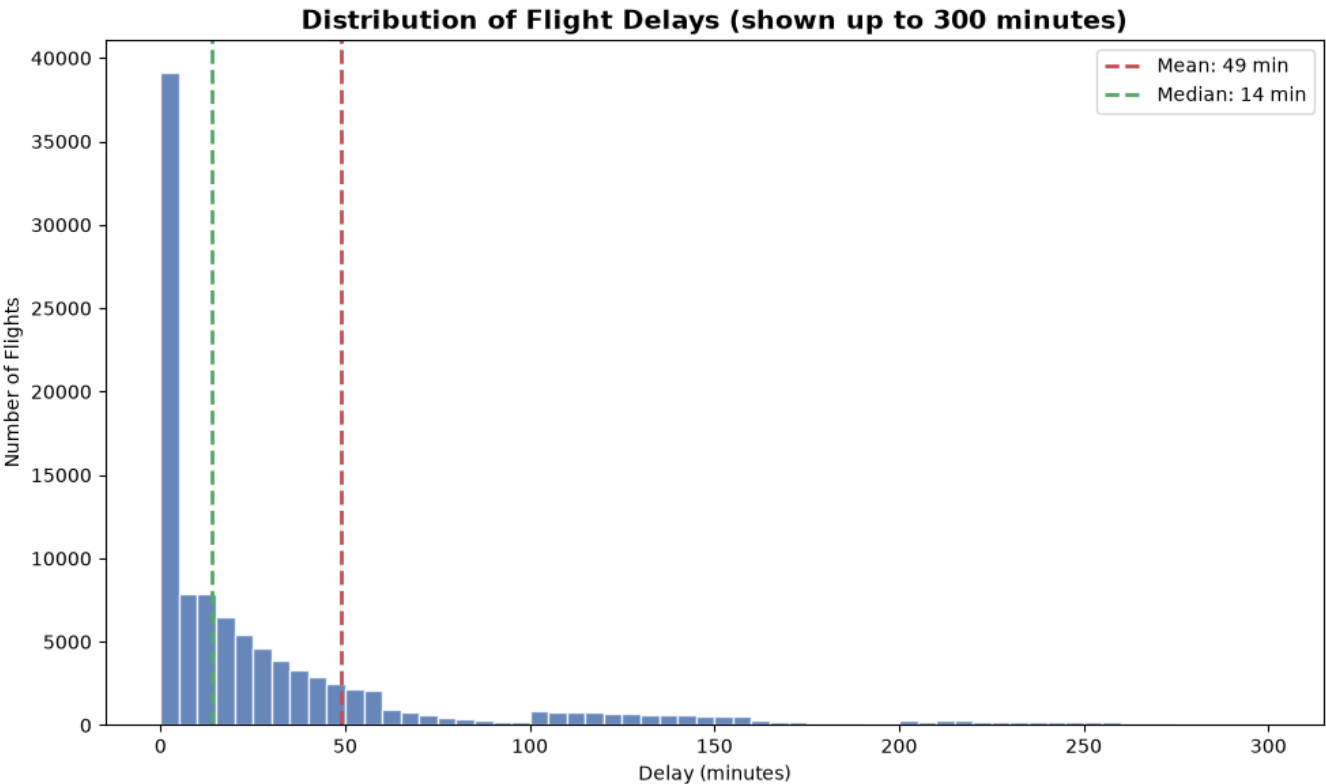
Experienced Some Delay

48.7 / 14

Mean / Median Delay (min)

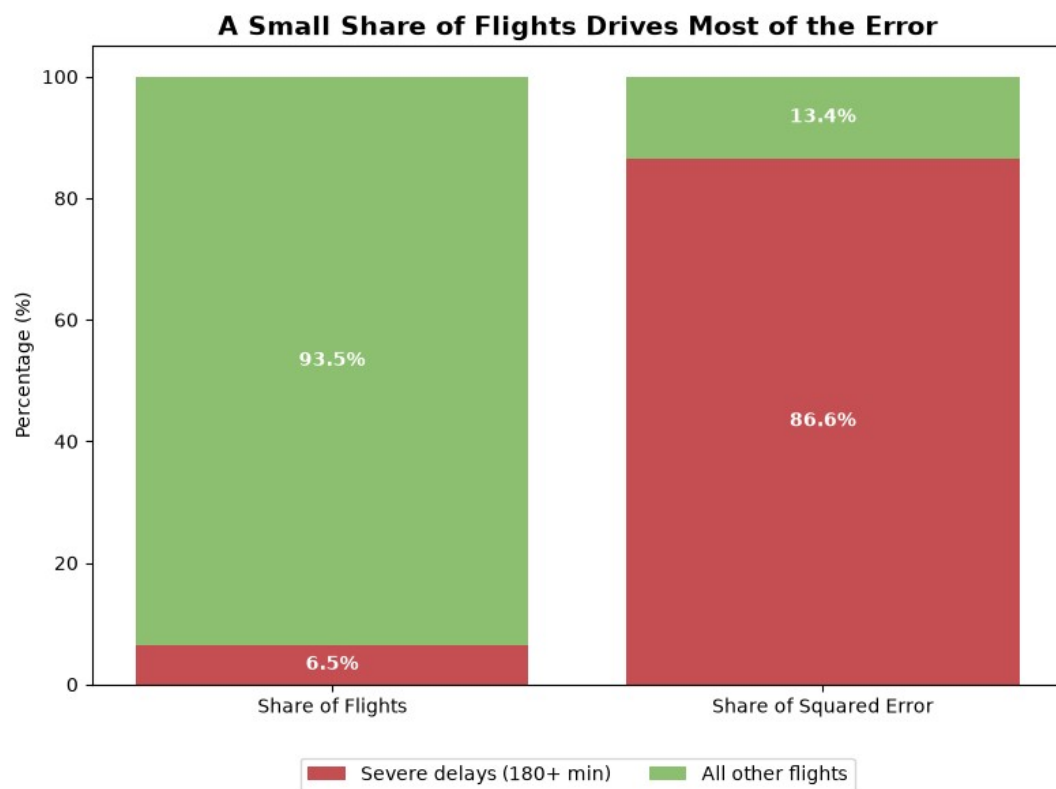
3,451

Max Delay Observed (min)



The distribution is strongly right-skewed: a long tail of extreme outliers pulls the mean well above the median — directly informing our metric choice, next.

A Small Share of Flights Drives Most of the Cost



6.5% of flights, 86.6% of the error

Severely delayed flights (180+ minutes) make up just 6.5% of the data, but account for 86.6% of total squared deviation — the exact quantity RMSE penalizes.

Why this matters for the business:

- A short delay is absorbed by schedule buffer at minimal cost.
- A long delay triggers crew reallocation, aircraft repositioning, and lost passenger trust.
- Cost scales non-linearly — a 3-hour delay is disproportionately worse than a 10-minute one.

RMSE (primary) — judges the model on rare, high-cost extremes.

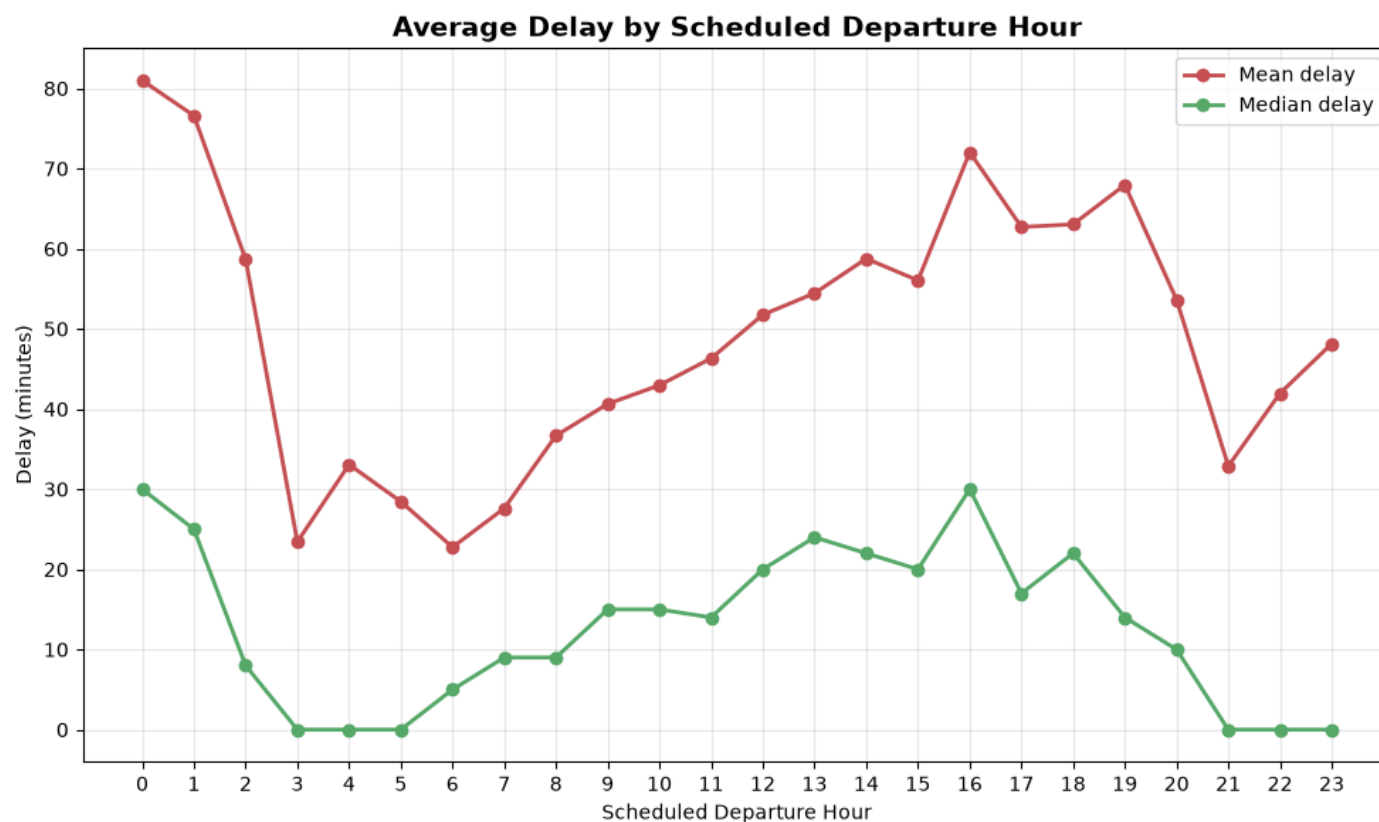
MAE (secondary) — reflects everyday operational accuracy.

From Raw Records to a Validated Model



Why time-based, not random, splitting? Several features (delay propagation, historical route/aircraft averages) are computed from the target variable and are time-dependent. A random split would let future flights leak information into a "past" flight's history — a time-based split makes this impossible by construction.

Delays Build Up Through the Day



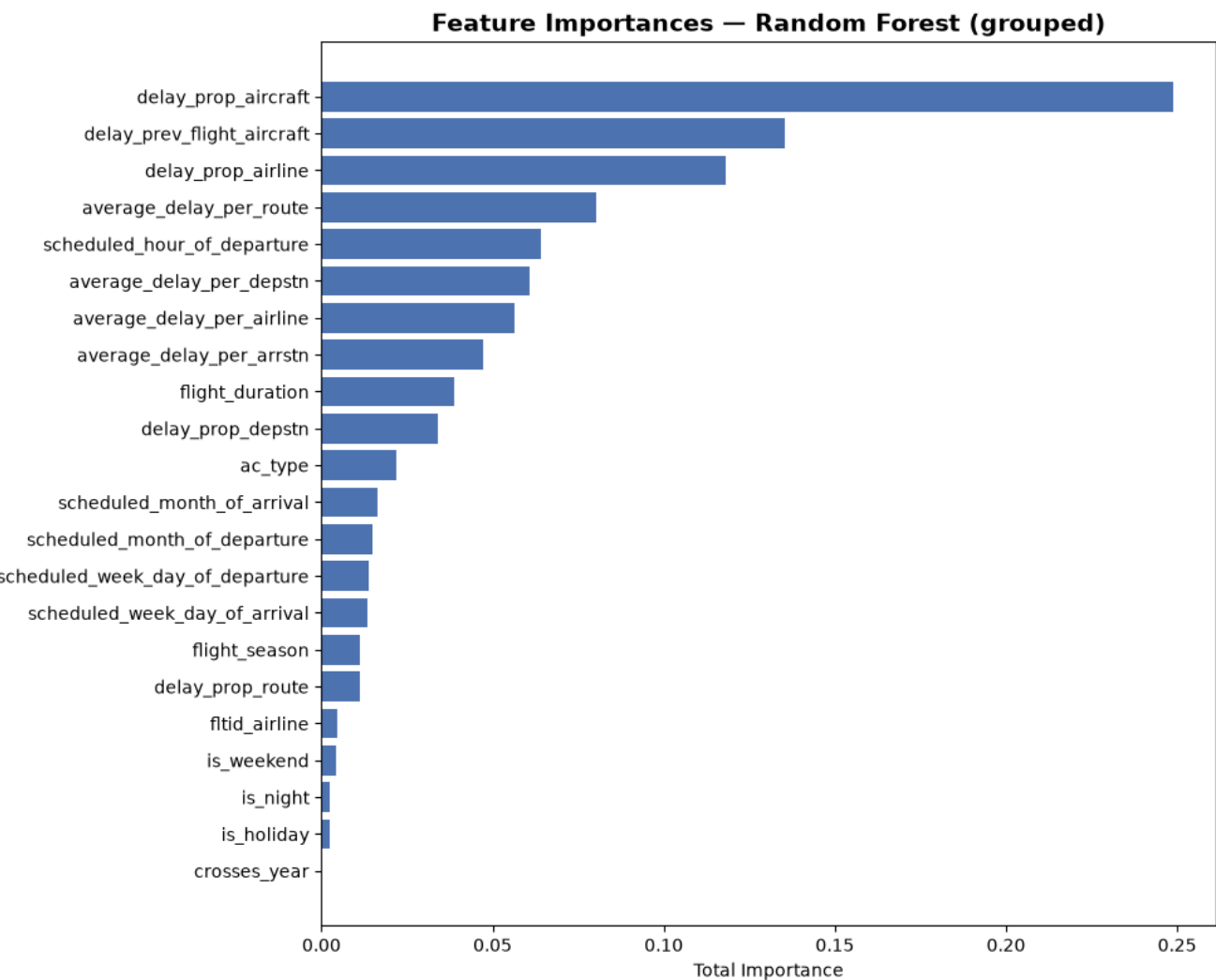
A clear, explainable pattern

Average delay is lowest overnight and climbs through the day as congestion and knock-on effects accumulate — exactly what "delay propagation" features are designed to capture.

15 engineered features, including:

- Same-day delay propagation (per aircraft, airline, route, airport)
- Delay of this aircraft's immediately preceding flight
- Historical average delay per route / airline / airport
- Flight duration, season, weekend & holiday flags

Real-Time Signals Beat Calendar Patterns



The top signal, by far

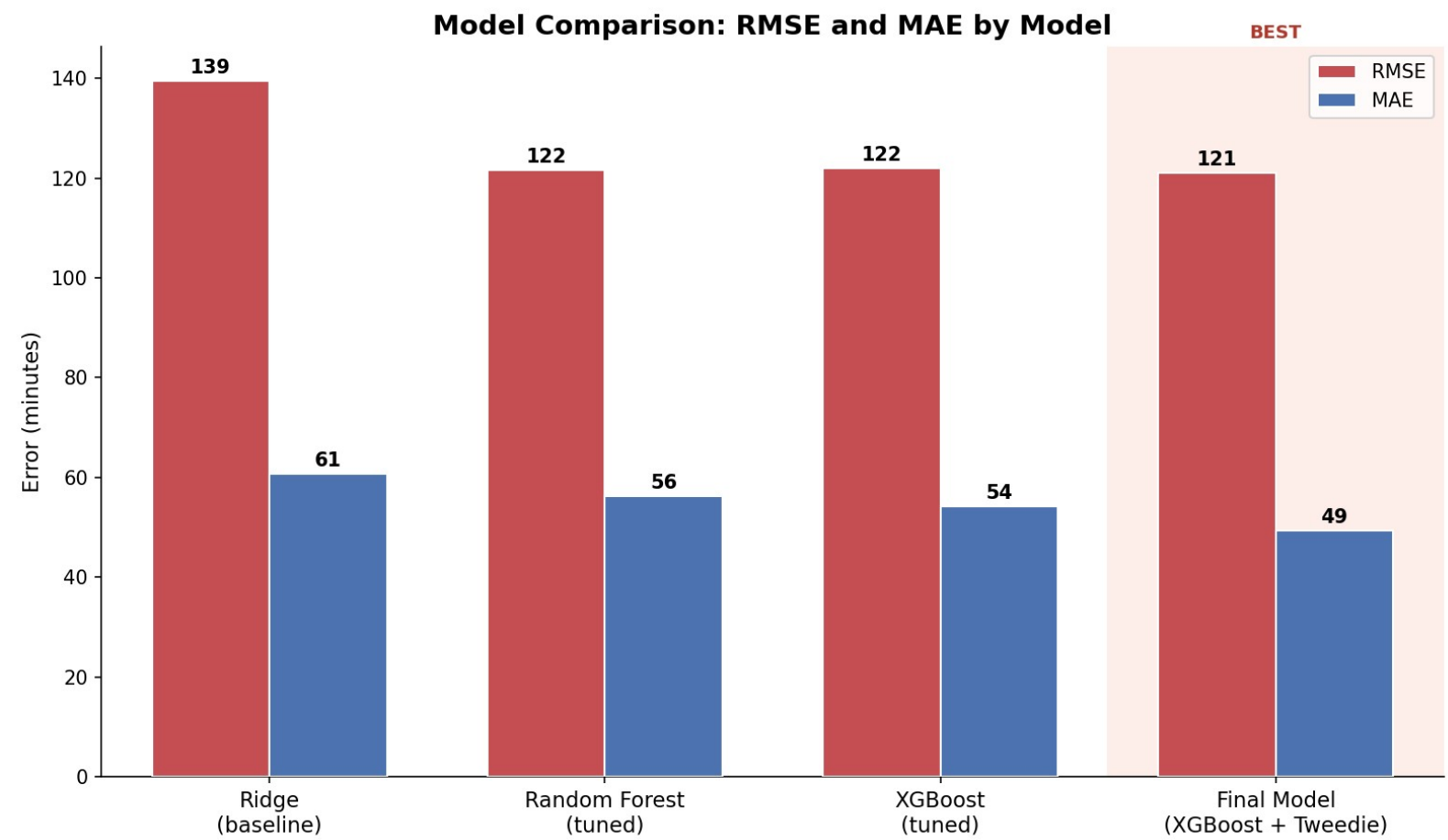
Whether this specific aircraft is already running late today is the single strongest predictor of delay — well ahead of route history, time of day, or aircraft type.

Key takeaway:

Delays are driven far more by “what's happening right now” than by “what usually happens on this route or season.” Calendar flags (holiday, weekend, night) added little once same-day propagation was included.

This finding directly shaped our modeling: we added a feature capturing each aircraft's immediately preceding flight delay, which measurably improved both RMSE and MAE.

A Clear, Compounding Improvement Over Baseline



-13.3%

RMSE vs. baseline

-18.6%

MAE vs. baseline

Final Model

XGBoost + Tweedie objective

Best RMSE **AND** best MAE across every model tested.

One Model, Best on Every Metric

XGBoost + Tweedie Objective

Best RMSE and best MAE simultaneously

121.0

RMSE

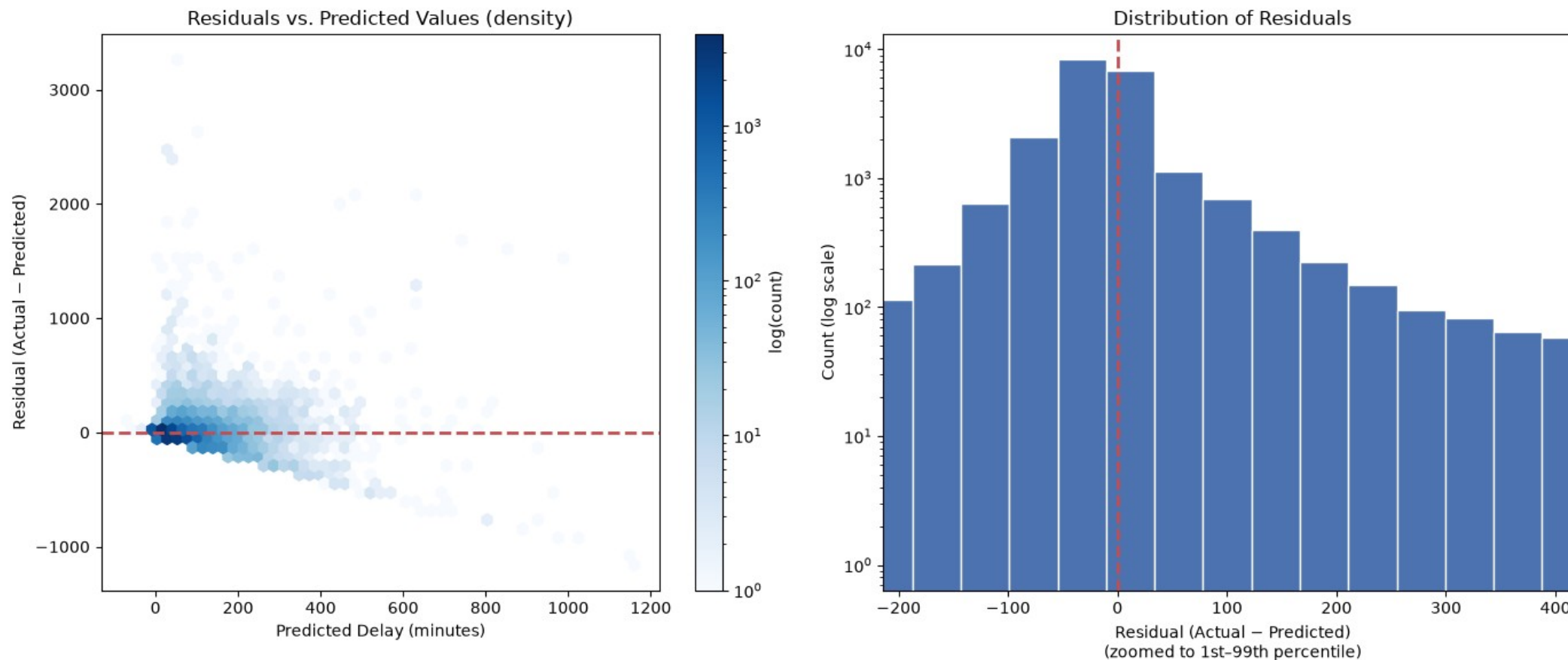
49.4

MAE

RECOMMENDED FOR PRODUCTION

Why it wins: the Tweedie objective is purpose-built for a non-negative, right-skewed target like delay minutes, and a rolling delay-propagation feature adds real-time signal — together improving RMSE by 13.3% and MAE by 18.6% over the Ridge baseline.

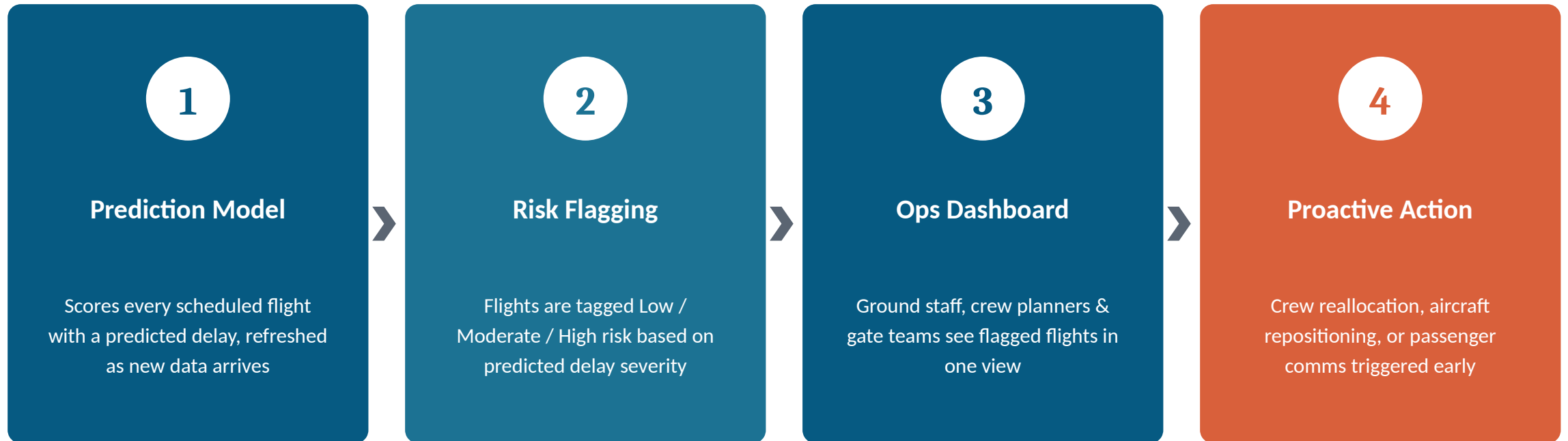
Well-Calibrated for the Vast Majority of Flights



Residual = actual delay minus predicted delay. The dense cluster near zero (left) and the tall spike at zero (right) show that predictions track the true delay closely for the large majority of flights. The model tends to under-predict the rare, most severe delays — the segment we prioritized RMSE specifically to hold the model accountable for.

A Delay-Risk Dashboard for Operations

Turning individual predictions into an operational early-warning tool:



Where We Push Further



Future Work

- Incorporate weather data — a likely driver of the severe-delay tail not yet captured.
- Specialized modeling for the high-delay-risk segment, layered on top of the current model.
- Extend rolling delay-propagation windows beyond the current 3-flight average.

Bottom line: the model already outperforms every baseline tested — these next steps would push it further, particularly on the hardest, most extreme delays.

Thank You

Questions & Discussion

Luca Tomarelli • Qutibah Hussein

Flight Delay Prediction | Tunisair Operational Data